

Abdelhamid Ezzer

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EDUCATION

UCL: University College London

Oct 2024 -

PhD in Statistics

London, United Kingdom

- I am a first-year PhD student supervised by Profs Jeremias Knoblauch and Ilija Bogunovic. My research interests lie in applying principled Generalised Bayesian methods to large-scale Machine Learning models.

Ecole Centrale Paris (currently known as CentraleSupélec)

Sep 2014 – Dec 2017

M.Sc in Applied Mathematics

Paris, France

- **Coursework:** Probability, Statistics, Numerical Methods, Partial Differential Equations, Functional Analysis, Programming...

ENS Paris-Saclay: Ecole Normale Supérieure Paris-Saclay

Sep 2016 – Dec 2017

Research M.Sc in Applied Mathematics and Machine Learning (MVA)

Paris, France

- **Coursework:** Deep learning, sparsity and compressed sensing, statistical learning, reinforcement learning...

University of Cambridge (Department of Engineering)

Oct 2015 – Jun 2016

Erasmus exchange program

Cambridge, United Kingdom

- **Coursework:** Software Engineering and Design, Computer and Network Systems, Signals and Systems, Thermodynamics...

EXPERIENCE

Fimatix

Jan 2024 – Aug 2024

Data Scientist

London, United Kingdom

- I worked on building an LLM-powered pipeline for extracting information from unstructured documents.
- **Key responsibilities:** Design, implement and evaluate models./ Deploy scalable models into production./ Decide the next research directions.
- **Keywords:** Deep Neural Networks, Natural Language Processing, Question Answering...

Amazon

Feb 2018 – Jun 2023

Applied Scientist

Cambridge, United Kingdom

- I worked as an Applied Scientist within Amazon's Alexa Text-to-Speech Research team. I contributed to several Alexa's features, such as celebrity TTS voices and accented speech.
- **Key responsibilities:** Design, implement and evaluate models./ Propose new research directions./ Efficiently report research findings through internal reports and external publications.
- **Keywords:** Deep Neural Networks, Speech Synthesis, sequence-to-sequence modelling, Variational Auto-Encoders, Generative Adversarial Networks, Normalizing Flows...

AXA Data Innovation Lab (DIL)

May 2017 – Oct 2017

Intern Data Scientist

Paris, France

- **Key responsibilities:** Design, implement and evaluate models.
- Implemented a graph-based model to detect insurance policy fraud. The graph-based model achieved significant improvements to the absolute accuracy score.

IBM

Oct 2016 – Mar 2017

Part-time Data Scientist

Paris, France

- Worked in a two-student group to construct a Machine Learning model capable of detecting credit card fraud. The proposed model significantly outperformed a rule-based baseline.

SKILLS

Languages: Arabic (native), French (native), English

Programming languages: Python, Scala, R, C++

Deep Learning Frameworks: Pytorch, MXNet

Tools: L^AT_EX, MS Office, Matlab, SQL

Relevant skills/knowledge: Probability, Statistics, Machine Learning, Deep Learning, Programming

Familiar Deep Learning concepts: sequence-to-sequence modelling (including Transformers), Variational Auto-Encoders, Generative Adversarial Networks, Normalizing Flows, Diffusion models...

PUBLICATIONS

- Jakub Mosinski, Piotr Bilinski, Thomas Merritt, **Abdelhamid Ezzer**, Daniel Korzekwa: *AE-FLOW: Autoencoder Normalizing Flow*. ICASSP 2023
- Guangyan Zhang, Thomas Merritt, Manuel Sam Ribeiro, Biel Tura Vecino, Kayoko Yanagisawa, Kamil Pokora, **Abdelhamid Ezzer**, Sebastian Cygert, Ammar Abbas, Piotr Bilinski, Roberto Barra-Chicote, Daniel Korzekwa, Jaime Lorenzo-Trueba: *Comparing normalizing flows and diffusion models for prosody and acoustic modelling in text-to-speech*. INTERSPEECH 2023.
- **Abdelhamid Ezzer**, Thomas Merritt, Kayoko Yanagisawa, Piotr Bilinski, Magdalena Proszewska, Kamil Pokora, Renard Korzeniowski, Roberto Barra-Chicote, Daniel Korzekwa. *Remap, warp and attend: Non-parallel many-to-many accent conversion with Normalizing Flows*. IEEE Speech Language Technology Workshop (SLT) 2022.
- Piotr Bilinski, Thomas Merritt, **Abdelhamid Ezzer**, Kamil Pokora, Sebastian Cygert, Kayoko Yanagisawa, Roberto Barra-Chicote, Daniel Korzekwa. *Creating New Voices using Normalizing Flows*. INTERSPEECH 2022.
- Magdalena Proszewska, Grzegorz Beringer, Daniel Sáez-Trigueros, Thomas Merritt, **Abdelhamid Ezzer**, Roberto Barra-Chicote. *GlowVC: Mel-spectrogram space disentangling model for language-independent text-free voice conversion*. INTERSPEECH 2022.
- Thomas Merritt, **Abdelhamid Ezzer**, Piotr Bilinski, Magdalena Proszewska, Kamil Pokora, Roberto Barra-Chicote, Daniel Korzekwa. *Text-free non-parallel many-to-many voice conversion using normalising flows*. International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2022.
- **Abdelhamid Ezzer**, Adam Gabrys, Bartosz Putrycz, Daniel Korzekwa, Daniel Saez-Trigueros, David McHardy, Kamil Pokora, Jakub Lachowicz, Jaime Lorenzo-Trueba, Viacheslav Klimkov. *Enhancing audio quality for expressive Neural Text-to-Speech*. Accepted for publication in the Speech Synthesis Workshop (SSW) 2021.
- Raahil Shah, Kamil Pokora, **Abdelhamid Ezzer**, Viacheslav Klimkov, Goeric Huybrechts, Bartosz Putrycz, Daniel Korzekwa, Thomas Merritt. *Non-Autoregressive TTS with Explicit Duration Modelling for Low-Resource Highly Expressive Speech*. Accepted for publication in the Speech Synthesis Workshop (SSW) 2021.

PATENTS

- "Voice customization for synthetic speech generation". US patent #12100383. Publication date: 2024/09/24
- "User-customized synthetic voice". US patent #12087270. Publication date: 2024/09/10

HONORS AND AWARDS

- French government's Major Excellence scholarship. I was awarded a scholarship for my admission to Ecole Centrale Paris.
- Ranked 2nd in the national Baccalaureate exams in Morocco in the Applied Mathematics B option.
- Ranked 5th in the national Olympiads of Physics, Morocco.